FUTURE CITY FELLOWSHIP · MILESTONE 2 · JULY 21, 2026

A Staged, Future-Ready Public EV Charging Plan

Fairfax City + Prince William County, Virginia — 12 candidate sites, 25 starter ports on a 48-port future-ready pathway

Team: Anna · Laylas · Andy · Adama · Paula

Target 13 minutes · static visuals only (all images open offline)

All model figures are screening-class outputs of canonical contract v1.0.0 (2026-07-19); nothing in this deck is a budget, permit, or client endorsement.

Build public charging that grows with demand — without repeating two local failures

Phase 1 committed us to equitable public Level 2 charging at municipal sites in Fairfax City and Prince William County.

Milestone 2 answers: which of 12 candidate sites, how many ports, on what staging logic, over 10 / 20 / 30 years (2036 / 2046 / 2056).

Failure mode 1 — retrofit cost escalation

Expansion pathways not built during the first dig force expensive trenching later (documented locally at McCoart).

Failure mode 2 — overbuild and underuse

Capacity installed ahead of demand sits idle and costs money to keep online or decommission (documented locally at Neabsco).

Scope: public AC Level 2 only — no DC fast charging, no battery storage at any of the 12 sites.

Basis: Phase 1 problem statement carried into Milestone 2 · client interview evidence (PWC, July 2026) · 12 candidate public sites

Two pieces of client evidence anchor the design

McCoart Government Center — trenching

“The cost driver was trenching distance — expansion pathways were not in the ground when the first ports went in.”

— [client contact] , Prince William County (paraphrased from July 2026 interview)

Neabsco commuter garage — overbuild

“70 ports went in; 17 stations were later taken offline — roughly $1,000 per station per year at stake.”

— [client contact] , Prince William County (paraphrased from July 2026 interview)

Also from the client record: ADA EV-space redesign cost about a year (Dominion reports the same); a ~35-station county fleet hub is in design (parallel track).

Fire / HazMat ([fire-service interviewee]): no local EV fires yet, but a responder recognition-and-education gap exists — and charger siting can block emergency access.

Evidence: recorded stakeholder interviews, July 2026 ([client contact] , PWC; [fire-service interviewee] , Fire/HazMat) · quotes paraphrased, attributed in the interview record

Strong client evidence in hand; the compliant Meeting #2 is prepared, not yet held

What we have

• A substantive July interview with [client contact] (PWC) plus fire-service evidence from [fire-service interviewee] — real client input that already changed the design.

What we do not yet have

• A fellowship-compliant Client Meeting #2: no dated session where the client reviewed our alternatives and gave recorded feedback on them.

What is prepared and ready to send

• A complete follow-up package — pre-read, agenda, plain-language alternatives briefing, feedback form, decision log — [client contact] offered a follow-up meeting or email exchange.

Claims discipline: nothing in this deck asserts that Client Meeting #2 occurred or that the client endorsed any alternative.

Artifacts: human_actions/client_meeting_2/ (pre-read, agenda, briefing, feedback form, what-changed register — register stays blank until the touchpoint)

04 · SEVEN GENUINELY DIFFERENT WAYS TO DO THIS

The alternatives we compared — including the honest “do nothing”

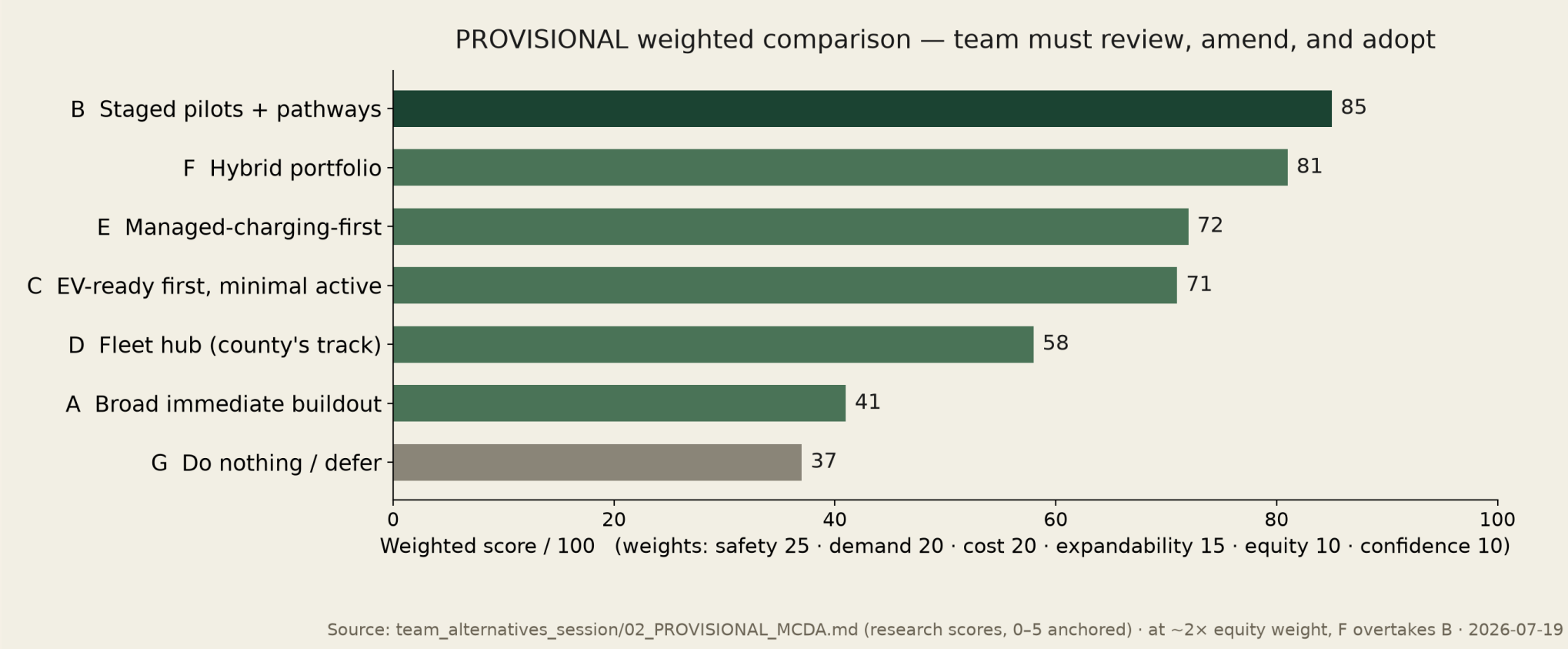
	Alternative	Year-1 picture	10-yr cost order	Main risk
A	Broad immediate buildout	Dozens of ports, mostly idle	Highest	Repeating Neabsco underuse
B	Staged pilots + expandable pathways	2–3 ports live at top sites; conduit oversized	Low–moderate	Slower geographic reach
C	EV-ready first (pathways only)	Capped conduit, little service	Lowest capital	No visible public value
D	Centralized fleet hub + selective public	County's ~35-station hub advances (its own track)	Moderate–high	ADA clustering cost; no duty-cycle data
E	Managed-charging-first	Smart charging on existing ports	Low–moderate	Needs networked EVSE + fail-safe comms
F	Hybrid portfolio (B+C+E, coord. with D)	Pilots plus EV-ready seeds	Slightly above B	More moving parts to govern
G	Do nothing / defer (comparator)	Nothing	Zero direct cost	Wastes the proactive planning window

The county fleet hub (D) is the client's real parallel track — we coordinate with it; it is not a competing recommendation.

Source: human_actions/client_meeting_2/04_ALTERNATIVES_BRIEFING.md (drafted 2026-07-19, not sent) · seven researched alternatives, none invented for show

05 · WEIGHTED COMPARISON — PROVISIONAL, PENDING TEAM ADOPTION

Staged distributed pilots lead at 85/100 — hybrid portfolio close behind at 81



Source: team_alternatives_session/02_PROVISIONAL_MCDA.md (research scores, 0–5 anchored) · at ~2× equity weight, F overtakes B · 2026-07-19

Why B wins: it is the only option supported by both the client's operating history (McCoart, Neabsco) and the model's low simulated utilization. Disclosed sensitivity: if equity is weighted ~2× the default, hybrid F overtakes — so we present B as the hybrid's first stage.

Source: team_alternatives_session/02_PROVISIONAL_MCDA.md — research scores; the team must review, amend, and adopt before this becomes the team's decision

Six steps, with hard gates before any design freeze or expansion

1 · Rank the 12 sites (model of record)

2 · Verify top sites: electrical, AHJ, ADA, fire access

3 · Pilot 2–3 ports per site; oversize conduit + panel during the dig

4 · Monitor with dual-path utilization logging

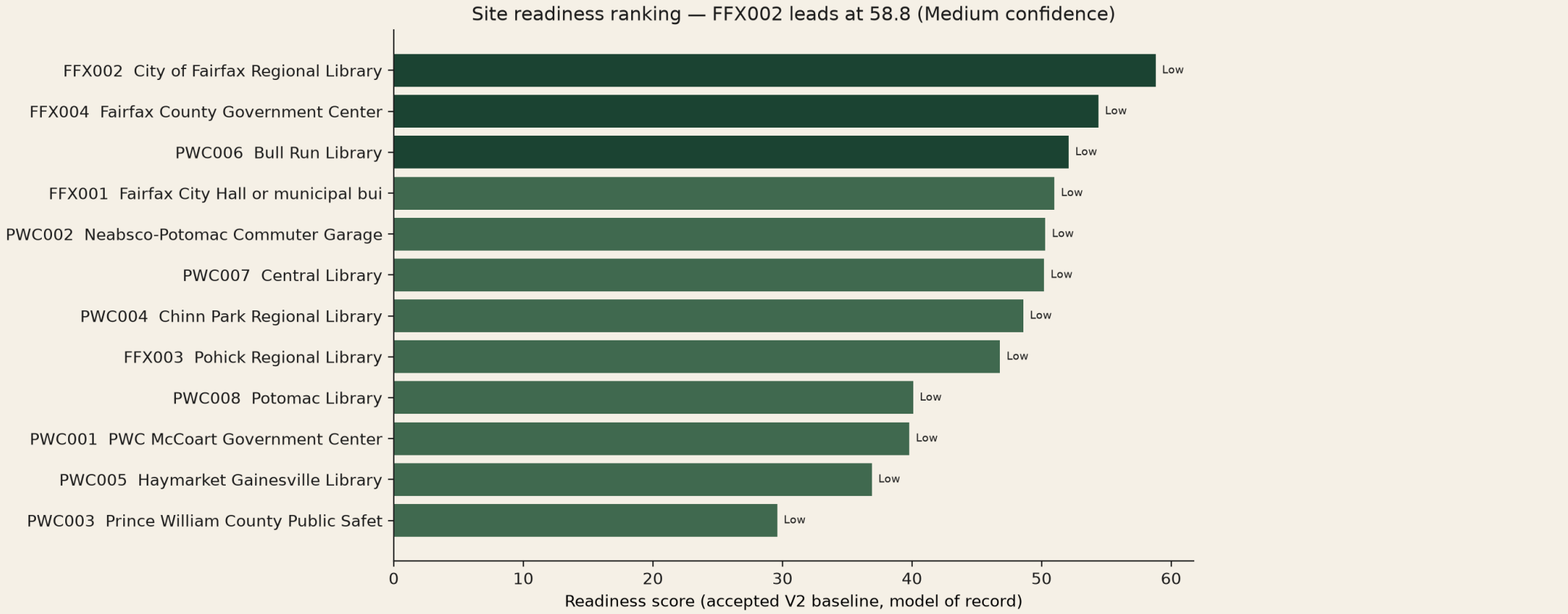
5 · Expansion trigger fires (utilization + equity override)

6 · Energize the next ports on the pre-built pathway

Gates that stop us: service-capacity confirmation (currently placeholder), AHJ permit interpretation (2020 NEC via 2021 VA USBC), ADA layout review, fire-access review, funding eligibility. No gate passed = no build.

Service capacity is a PLACEHOLDER at 12 of 12 sites — utility / facility confirmation pending; no design freeze until confirmed.

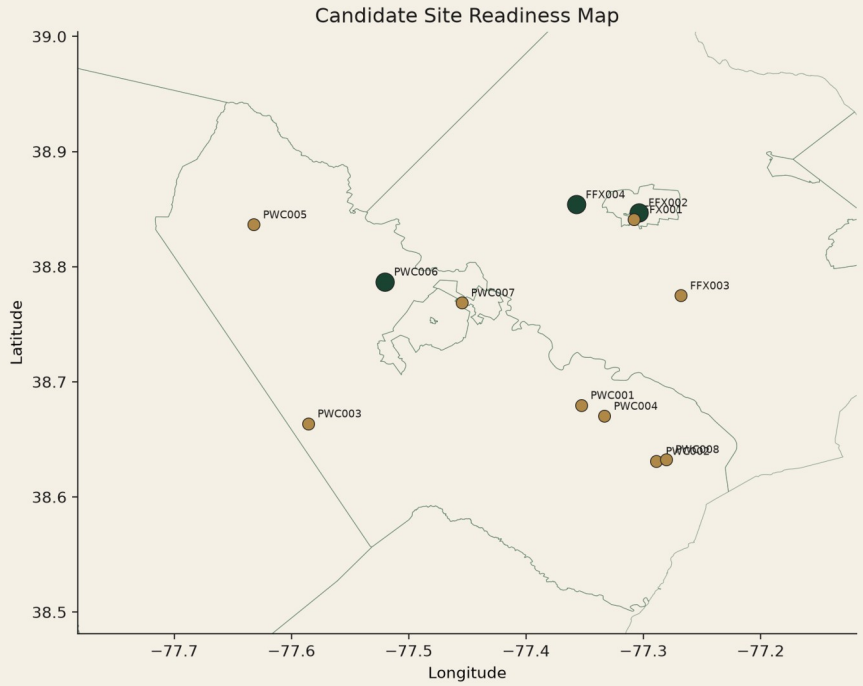
FFX002 City of Fairfax Regional Library leads at 58.8 — no site earns High data confidence



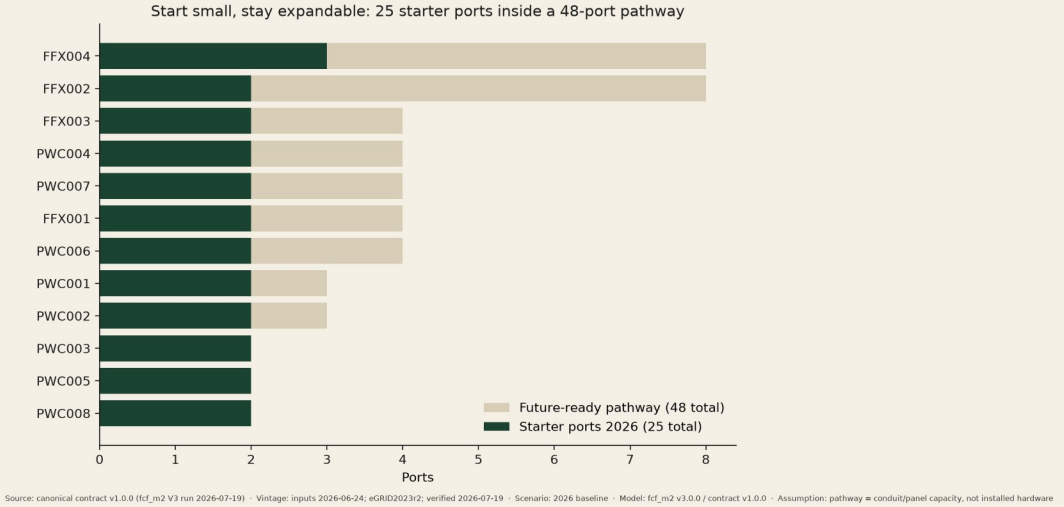
Top three: FFX002 58.8 (Medium overall confidence) · FFX004 Fairfax County Government Center 54.4 · PWC006 Bull Run Library 52.1 (top PWC site). Per-site data confidence is Low at all 12 sites — 38 REVIEW + 12 INFO findings stand open.

Source: canonical data contract v1.0.0 (fcf_m2 v3 run 2026-07-19) · inputs 2026-06-24 · screening-class values, not budgets · eGRID2023 Rev 2 SRVC 596.3 lb CO2e/MWh, accessed 2026-07-19

12 sites · 25 starter ports · 48-port future-ready pathway



Source: US Census TIGER/Line 2024 (cached in repo) + real_sites_v2.csv · Vintage: TIGER 2024 / sites 2026-06-24 · Scenario: site map, offline (FCF_OFFLINE set) · Model: fcf_m2 v3.0.0 / contract v1.0.0 · Assumption: no basemap tiles in offline mode



Source: canonical contract v1.0.0 (fcf_m2 V3 run 2026-07-19) · Vintage: inputs 2026-06-24; eGRID2023r2; verified 2026-07-19 · Scenario: 2026 baseline · Model: fcf_m2 v3.0.0 / contract v1.0.0 · Assumption: pathway ≠ conduit/panel capacity, not installed hardware

Starter ports serve today's demand; the pathway (conduit + panel space) is built once, during the first dig — the direct answer to the McCoart trenching evidence.

Service capacity is a PLACEHOLDER at 12 of 12 sites — utility / facility confirmation pending; no design freeze until confirmed.

Source: canonical data contract v1.0.0 (fcf_m2 v3 run 2026-07-19) · inputs 2026-06-24 · screening-class values, not budgets · eGRID2023 Rev 2 SRVC 596.3 lb CO2e/MWh, accessed 2026-07-19 · offline map, no live tiles

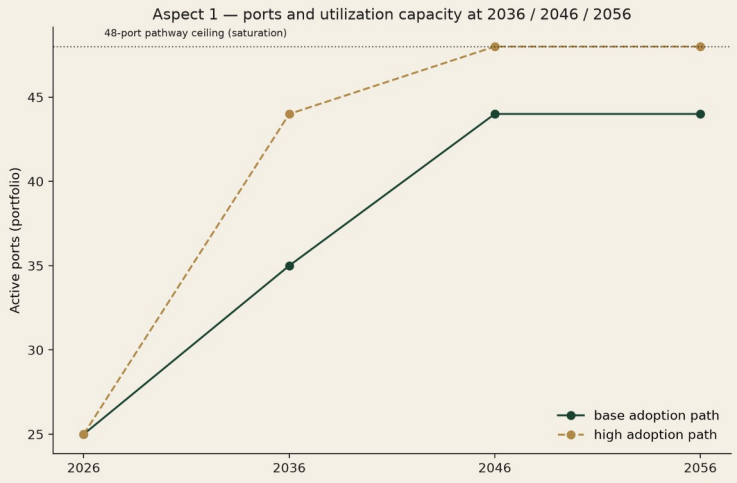
One table: demand/ports and lifecycle cost at 2036 · 2046 · 2056

Metric (portfolio of 12 sites)	2026	2036	2046	2056
Aspect 1 — active ports, base path	25	44	47	47
Aspect 1 — active ports, high path	25	47	48	48
Future-ready pathway capacity (built once)	48	48	48	48
Aspect 2 — NPC, staged build (cumulative)	—	$1.56M	$2.29M	$3.01M
Aspect 2 — NPC, broad day-1 build	—	$1.87M	$2.64M	$3.41M
Avoided overbuild cost (staged vs broad)	—	$0.31M	$0.34M	$0.39M
Net avoided CO2e, t/yr (base / high)	≈43	≈82 / 113	≈130 / 195	≈173 / 260

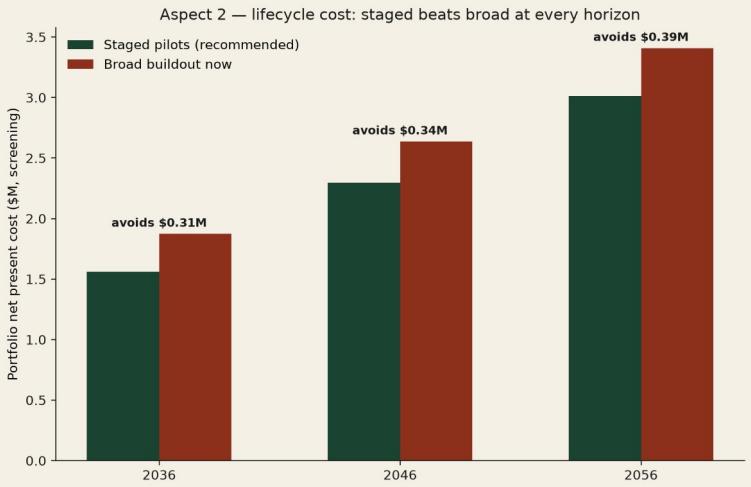
Service capacity is a PLACEHOLDER at 12 of 12 sites — utility / facility confirmation pending; no design freeze until confirmed. Costs are screening-class net present cost (base escalation), never budgets.

Source: canonical data contract v1.0.0 (fcf_m2 v3 run 2026-07-19) · inputs 2026-06-24 · screening-class values, not budgets · eGRID2023 Rev 2 SRVC 596.3 lb CO2e/MWh, accessed 2026-07-19

Ports track demand; staged spending stays below broad buildout



Source: canonical contract v1.0.0 (fcf_m2 V3 run 2026-07-19) · Vintage: inputs 2026-06-24; eGRID2023r2; verified 2026-07-19 · Scenario: V3 canonical horizons · Model: fcf_m2 v3.0.0 / contract v1.0.0 · Assumption: growth multipliers provisional (uncalibrated); trigger-gated activation



Source: canonical contract v1.0.0 (fcf_m2 V3 run 2026-07-19) · Vintage: inputs 2026-06-24; eGRID2023r2; verified 2026-07-19 · Scenario: base escalation, 3% real discount · Model: fcf_m2 v3.0.0 / contract v1.0.0 · Assumption: screening-class benchmarks — not budgets or quotes

Left — Aspect 1: active ports climb the 48-port pathway as adoption grows (base vs high path). Right — Aspect 2: staged net present cost stays $0.31–0.39M below broad day-1 buildout at every horizon.

Service capacity is a PLACEHOLDER at 12 of 12 sites — utility / facility confirmation pending; no design freeze until confirmed.

Source: canonical data contract v1.0.0 (fcf_m2 v3 run 2026-07-19) · inputs 2026-06-24 · screening-class values, not budgets · eGRID2023 Rev 2 SRVC 596.3 lb CO2e/MWh, accessed 2026-07-19

Two highlighted ULSE standards, the law as adopted, and one proposed standard

Highlighted (ULSE)

UL 2594 (3rd ed., 2022) — safety for the exact AC L2 equipment class proposed.

UL 3600 (2023) — circular-economy measurement; procurement disclosure + annual circularity reporting.

Supporting / law

UL 2231-1/-2 personnel protection (in the EVSE listing).

2020 NEC Art. 625 via 2021 VA USBC (eff. 2024-01-18) — the 2023 NEC is NOT Virginia law.

Access Board EVSE accessibility rule: still at NPRM — design to TA guidance now.

Conditional — NOT active

UL 2202 (DC), UL 9540/9540A + NFPA 855 (storage) apply only if DCFC or batteries enter scope.

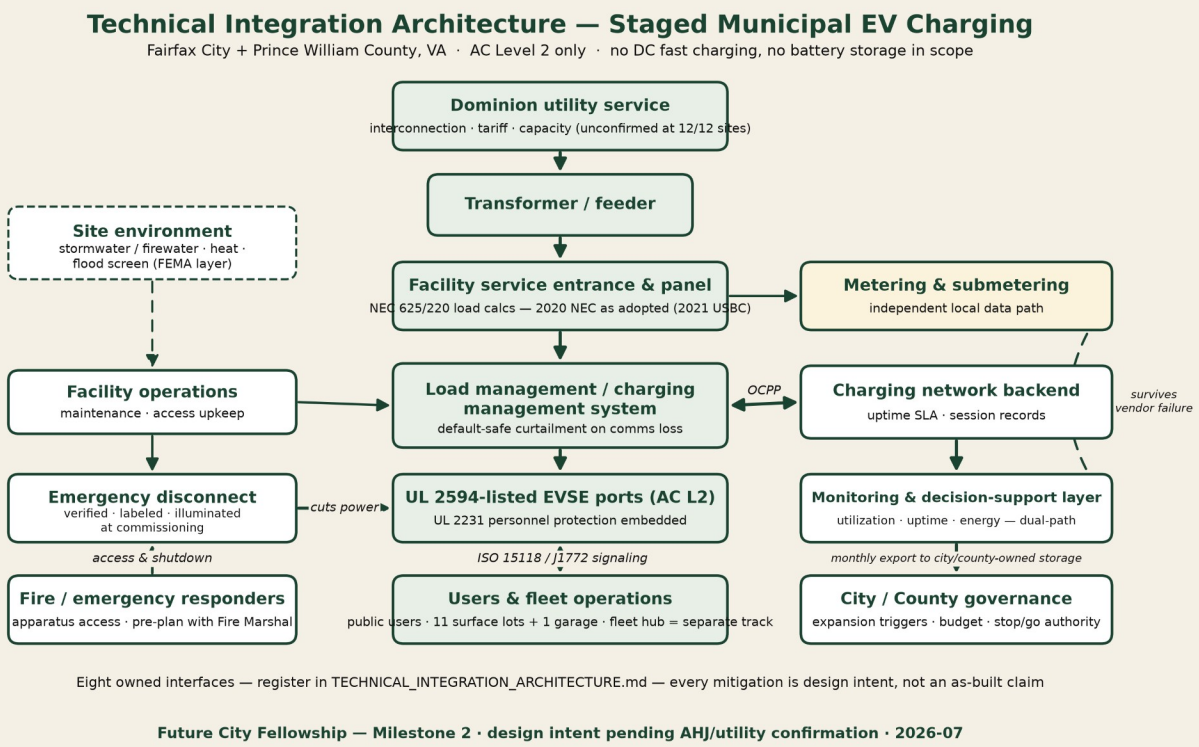
Today: no DC fast charging, no battery storage at any of the 12 sites.

Proposed standard (fellowship design concept — not an existing or adopted standard):

“Municipal EV Charging Site Readiness and Emergency Access Standard” — modules: emergency access & disconnect · expandability / right-sizing documentation · accessibility cross-references.

Source: FINAL_STANDARDS_APPLICATION_MATRIX.md (verified 2026-07-19) · design intent only — all permit determinations rest with the AHJ

One architecture, three client-resonant failure modes, three mitigations



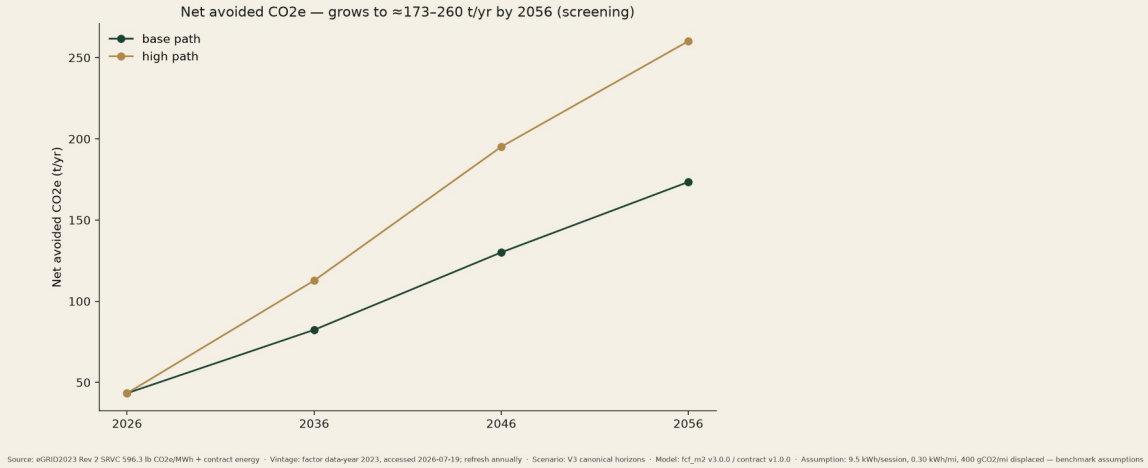
Top-3 failure modes (each one the client has lived):

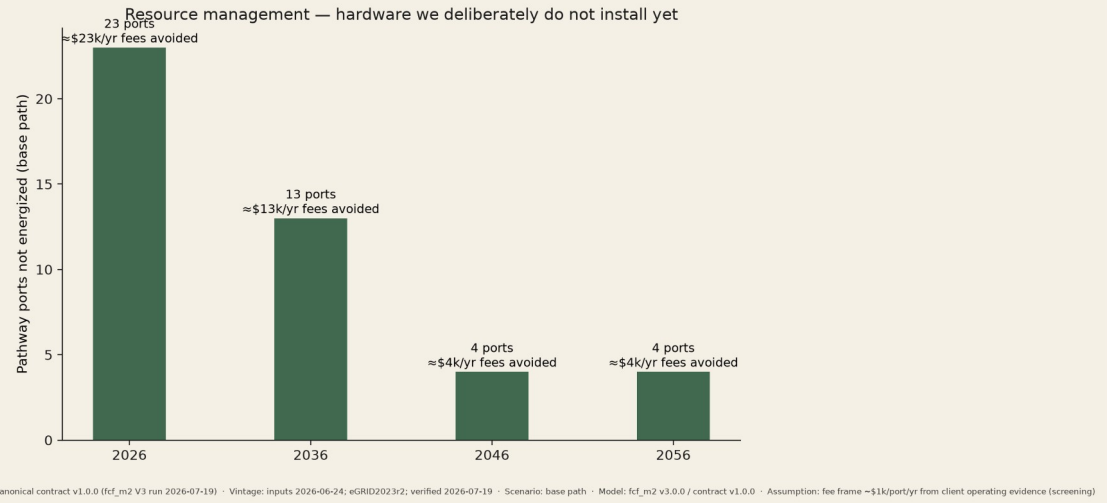
- Overbuild / underuse — Neabsco. Mitigation: staged triggers + dual-path utilization logging.
- Trenching cost blowout — McCoart. Mitigation: service-capacity confirmation before design freeze; pathways oversized in the first dig.
- Unclear emergency shutdown — fire-service evidence. Mitigation: verified, labeled, illuminated disconnect at commissioning + fire-department walkthrough.

Comms loss defaults to safe curtailment — hardwired overcurrent protection never depends on software.

Source: TECHNICAL_INTEGRATION_ARCHITECTURE.md + RISK_REGISTER.md (15 assessed failure modes) · every mitigation is design intent pending AHJ/utility confirmation

Avoided overbuild, growing CO2e benefit, and a realistic funding path





Funding: Dominion Level 2 Charging Program — 50% upfront make-ready, 100% in disadvantaged communities (eligibility/queue to confirm). NEVI is volatile (2025 freeze → court-ordered release → new guidance); treat strictly as upside, never the base. Library and commuter sites put ports where daily civic traffic already is; each site carries an equity indicator.

Source: canonical data contract v1.0.0 (fcf_m2 v3 run 2026-07-19) · inputs 2026-06-24 · screening-class values, not budgets · eGRID2023 Rev 2 SRVC 596.3 lb CO2e/MWh, accessed 2026-07-19 · no funding is awarded, applied for, or implied

The model is a screening tool; humans must validate before anyone builds

- 0 of 12 sites reach High data confidence; 38 REVIEW + 12 INFO findings remain open.
- Service capacity is a placeholder at 12/12 sites — utility / facility confirmation pending.
- Demand is synthetic and uncalibrated; costs are screening-class, never budgets.
- The MCDA is provisional until the team adopts it; Client Meeting #2 follow-up not yet held.
- Human validation still required: utility, facility managers, AHJ / electrical inspector, fire marshal, ADA review.

What this analysis is NOT

- 0 of 12 sites have High data confidence — every electrical value awaits confirmation
- Capacity is a PLACEHOLDER at 12/12 sites (no utility/facility verification yet)
- Demand model is synthetic and uncalibrated; growth multipliers are provisional
- Costs are screening-class benchmarks — not budgets, quotes, or engineering estimates
- No site is approved: electrical, ADA, fire, stormwater, and AHJ gates all remain
- Client Meeting #2 follow-up is prepared but not yet held — nothing implies endorsement
- No MATLAB simulation results are claimed (no licensed execution available)

Why we show this slide

Honest uncertainty is the evidence discipline that makes the staged, trigger-gated strategy credible: verify first, energize when data says so.

Source: canonical contract v1.0.0 (fcf_m2 V3 run 2026-07-19) · Vintage: inputs 2026-06-24; eGRID2023r2; verified 2026-07-19 · Scenario: limitations panel · Model: fcf_m2 v3.0.0 / contract v1.0.0

Source: contract v1.0.0 confidence fields + verification findings register (2026-07-19)

One question we genuinely need help with

“How should a city set an equitable expansion trigger for public EV charging when early utilization data are sparse — so it avoids both overbuilding and under-serving communities whose current EV adoption may be lower?”

Immediate next actions (this week)

- Send the utility / facility data requests for the top-3 sites (retire the capacity placeholder).
- Send the prepared Client Meeting #2 follow-up package — [client contact] offered a meeting or email exchange.
- Team session: adopt (or amend) the provisional MCDA and assign Paula's presentation role.
- Pilot ask: approve 2–3 starter ports at FFX002 / FFX004 / PWC006, screened against the Dominion Level 2 program.

Peer question ranked 95/100 in our question MCDA; two backup questions are ready in the Q&A pack

CLOSING

Speaking roster — July 21, 2026 · target 13:00

Speaker	Sections (beats)	Slides
Anna	Opening, problem, economics, peer question, close	1–2, 14, 16–17
Laylas	Client evidence, CM2 status, standards, limitations	3–4, 12, 15
Adama	Alternatives, weighted comparison, three-horizon results	5–6, 10–11
Andy	Solution workflow, model results, integration & risk	7–9, 13
Paula	TO BE ASSIGNED at Monday rehearsal — edit this row before presenting	—

Rehearsals: Monday ~8 pm (full run, timed) and Tuesday pre-meeting (cuts drill). Static visuals only — every image in this deck opens offline from presentation/static_visuals/.

Thank you — questions welcome; a Q&A backup pack accompanies this deck.